Fiscal Year:	FY 2025	**Task Last Updated:**	FY 06/13/2025
PI Name:	Weber, Richard Ph.D.		
Project Title:	The Origin of Fragility in High-Temperature Oxide Liquids		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Arlington Heights	**State:**	IL
Zip Code:	60004-7708	**Congressional District:**	8
Comments:	NOTE: Formerly with Containerless Research, Inc., until 2006.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Japanese Space Agency (JAXA) AO for Electrostatic Levitation Furnace (ELF) experiments on ISS
Start Date:	08/15/2019	**End Date:**	09/30/2025
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 9/30/2025 per NSSC information (Ed., 10/9/24) NOTE: End date changed to 9/30/2024 per NSSC information (Ed., 3/5/24) NOTE: End date changed to 9/30/2023 per NSSC information (Ed., 6/20/23) NOTE: End date changed to 9/30/2022 per M. Sansoucie/MSFC (Ed., 1/28/21) June 2019: Note project had preliminary work performed Feb-August 2019		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K1288 ; 80NSSC19K0518		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project is the United States (US) portion of an international Japan Aerospace Exploration Agency (JAXA) program to investigate the properties and structure of molten and supercooled metal oxides using the Electrostatic Levitation Furnace (ELF) on the International Space Station (ISS) and ground-based synchrotron x-ray and neutron studies. Four prototype compositions have been levitated and melted on ISS. Measurements of thermophysical properties, density, thermal expansion coefficient, viscosity and surface tension were made over a wide temperature range. Samples, including glasses, were recovered for ground based analysis and comparison with materials made on Earth. The work underpins development of a materials property-structure data set that can help in development of new materials and in meeting requirements of material utilization and processing in space. Additional samples are being prepared for measurements on ISS in 2023. Work is being published to provide data to the broader scientific community. The project team brings together the skills and experience needed to implement a complex research program. Reference: [1] S. Kohara, et al., “The origin of fragility in high-temperature oxide liquids - towards fabrication of novel non-equilibrium oxides,” JAXA flight project funded in 2017.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Developing an advanced understanding of the properties and structure of liquids that can be transformed into glasses is a high value enterprise that benefits the $300B global glass materials industry. Specific advances expected from this research will impact development of functional glasses by providing high accuracy thermophysical property and atomic structure data that can be used to optimize models used to design new materials from first principles. The titanate liquids being studied have high value applications. The compositions also relate to lunar mare regolith that contains relatively large amounts of titanium oxide. The work has led to discovery a new type of glass with highly coordinated titanium ions. Work is/has been published and a US patent on an optical device glass was filed.
	This project is part of a collaborative international JAXA/NASA program to investigate: (i) properties of liquid phase and supercooled metal oxides using the Electrostatic Levitation Furnace (ELF) on the International Space Station (ISS) [1,2], and (ii) atomic structure using ground-based synchrotron and neutron studies. We, the NASA investigators, are supplying samples for experiments using the ISS ELF to provide some of the first results from the instrument for US investigators. Samples were launched on SpaceX CRS-20 (SpX-20) in spring of 2020, on SpX-22 in June 2021, and on SpX-27 in March 2023. A holder was run in February 2024 with >80% success rate in melting levitated samples. In addition to several cited publications, the results are establishing a new class of optical glass materials and have led to new technology that is being patented. The project will be completed in 2025. A final report and other documents will be submitted as required under the terms of the contract. 1. STATUS: The project team includes Dr. Stephen Wilke; Mr. Abdulrahman Al-Rubkhi and Mr. Jared Rafferty from Materials Development Inc. (MDI); Dr. Shinji Kohara, National Institute for Materials Science, Japan; and Dr. Takehiko Ishikawa, Japanese Aerospace Exploration Agency (JAXA), Japan. The collaborators from Japan are not funded from this grant. The funded project work was completed in 2025. Some final in-progress publication edits are being completed. Data are being uploaded to the NASA Physical Sciences Informatics (PSI) system. 2. TECHNICAL ACTIVITIES: Activities during the reporting period included: (i) Analysis of data from flight experiments, (ii) Performing ground-based measurements on samples recovered from the flight experiments, and (iii) Publishing/presenting results and uploading data to the NASA PSI system. (i) A total of 30 samples covering 7 different compositions were investigated. For four compositions, multiple samples were stably levitated and fully melted. Drop size and oscillation power spectra were measured, and samples (glass for one composition) were recovered. Sample composition RT-3 was very successful for several reasons. It levitated reliably and could be stably held over a wide range of temperatures as both solid and liquid. This enabled detailed measurements of properties and investigation of glass formation in microgravity. The samples were suitable for very detailed measurement of atomic structure using x-ray and isotope substitution neutron diffraction and modelling [3]. The results led to important new insights into glasses that form from liquids that do not contain conventional network forming species. In addition, the critical cooling rate for glass formation for the RT-3 material is in a range that is easily accessible on ISS and on the ground so that samples could be made, compared, and contrasted. Analyses were performed on the video images, drop oscillation power spectra, and temperature data to obtain the temperature-dependent properties of the liquids. Considerable effort was invested into developing procedures for the analysis and to enable precise synchronization of the temperature and other measured data. A paper that describes the methods was published [4]. [Ed. Note: See References (below) and the Cumulative Bibliography.] Results of density, thermal expansion coefficient, surface tension and viscosity were published for several compositions. Please see cited references [4,5]. (ii) Ground based experiments were performed in several areas. These include: 1. Optical property measurements, including fluorescence and Raman. 2. Thermodynamic property measurements on samples made on the ground and on the ISS. The optical properties of several glasses have been measured in a collaboration with Professor Brian Topper at Clemson University and Professor Doris Moncke at Alfred University. The octahedral titanate network provides a low phonon energy host suitable for optical device materials. In addition, the glasses are strong and stable compared to typical low phonon energy halides. This work has led to four publications [6-10] and a patent filing [11]. Thermodynamic measurements on the glasses were made through a collaboration with Professor Alexandra Navrotsky, Dr. Tamil Subramani, and Dr. Laura Bonatti at the Navrotsky Eyring Center for Materials of the Universe (MotU) at Arizona State University [12]. The data show that the values of glass transition temperature and heat capacity of glasses made on Earth and in microgravity are within experimental error of each other.

Task Progress:	Glass formation in space occurred at a slightly smaller cooling rate than was possible on Earth. We do not yet have a full explanation for the reasons. The current hypothesis is that it is due to heat transfer effects that favor glass formation in quiescent liquids compared to the stirred liquids that occur on Earth. This result was the basis for a patent that was filed in 2024 [13]. (iii) Publishing/presenting results. Several papers have been published based on the ISS experiments. These are listed in the references section. Talks were presented at the American Society for Gravitational and Space Research (ASGSR) meeting in Puerto Rico, December 2024 (Weber); American Ceramic Society Glass and Optical Materials meeting in Vancouver, May 2025 (Wilke and Weber); and the 17th International Conference on the Physics of Non-Crystalline Solids (PNCS-17), Tsukuba, Japan, August, 2025 (Weber, Invited). Raw data is being uploaded to the NASA PSI system. Due to the large amount of video data from the drop shape and size measurements, the space allocation of the PSI account for this project was increased. Plans: The project work is completed. Some follow-up work is needed to complete final in-process publications. 3. REFERENCES CITED: [1] https:// [2] R. Weber, S.K. Wilke, and C.J. Benmore, “Containerless Techniques for in-situ X-Ray Measurements on Materials in Extreme Conditions,” J. Phys. Soc. Jpn., 91, 091008 (2022). [3] S.K. Wilke, O.L.G. Alderman, C.J. Benmore, J. Neuefeind and R. Weber, “Octahedral oxide glass network in ambient pressure neodymium titanate,” Sci. Rep., 12:8258 (2022), https:// [4] S.K. Wilke, A. Al-Rubkhi, C. Koyama, T. Ishikawa, H. Oda, B. Topper, E.M. Tsekrekas, D. Möncke, O.L.G. Alderman, V. Menon, J. Rafferty, E. Clark, A.L. Kastengren, C.J. Benmore, J. Ilavsky, J. Neuefeind, S. Kohara, M. SanSoucie, B. Phillips and R. Weber, “Microgravity effects on nonequilibrium melt processing of neodymium titanate: thermophysical properties, atomic structure, glass formation and crystallization,” npj Microgravity, 10:26, (2024). [5] S.K. Wilke, A. Alrubkhi, V. Menon, J. Rafferty, C. Koyama, T. Ishikawa, H. Oda, R. Hyers, R. Bradshaw, A. Kastengren, S. Kohara, M. SanSoucie, B. Phillips, and R. Weber, “Measuring the density, viscosity, and surface tension of molten titanates using electrostatic levitation in microgravity”, Appl. Phys. Lett., 124, 264102 (2024). [6] B. Topper, S.K. Wilke, M. Pettes, A. Alrubkhi, V. Menon, A. Neumann, D. Möncke, R. Weber, and A. Mafi, “Mid-infrared luminescence properties of erbium and dysprosium doped lanthanum titanate glasses,” Opt. Mats. Express, 13, 2857 (2023). [7] B. Topper, A. Neumann, A. Mafi, M. Pettes, A. Alrubkhi, S.K. Wilke and R. Weber, “Nonlinear optical properties of lanthanum titanate glasses prepared by levitation melting,” Appl. Phys. Lett., 124, 094104 (2024). [8] B. Topper, A. Neumann, S.K. Wilke, R.E. Youngman, A. Abdulrahman and R. Weber, “Comparing Pr3+ and Nd3+ for deactivating the Er3+: 4I13/2 level in lanthanum titanate glass,” Opt. Mats. Express, 14, 1309 (2024). [9] B. Topper, A. Neumann, S.K. Wilke, A. Alrubkhi, A. Mafi and R. Weber, “Site-selective fluorescence and spectroscopic properties of Yb-doped lanthanum titanate glasses,” Int. J. Glass. Sci., 15, 256 (2024). [10] J. Tolliver, S.K. Wilke, A. Alrubkhi, A. Neumann, G. Balakrishnan, R. Weber, and Brian Topper, “Nd-doped lanthanum titanate glass laser,” Opt. Components and Mats. XXII, S. Jiang & M.J.F. Digonnet, Eds. Proc. of SPIE, 13362, 1336204 (2025). https:// 10.1117/12.3042163 [11] B. Topper, A. Mafi, J.R. Weber, S.K. Wilke, Rare earth doped glasses and their application, US20240279109A1 (2024). [12] L. Bonatti, T. Subramani, S.K. Wilke, R. Weber, A. Navrotsky, Energetics of neodymium titanate glass made on Earth and in space, ACS Earth and Space Chemistry (2025), in press. [13] J.R. Weber, S.K. Wilke, Method for glass formation and morphology of products made from non-equilibrium liquids by processing in reduced gravity, US20250145513A1 (2024).
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